Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (3)
The Common-source Amplifier

Spring 2023

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The Common-source Amplifier

Objectives

- To study the operation of a common-source amplifier.
- To verify the voltage gain relations of the common-source amplifier.

Pre-lab Work

Find general expressions for the voltage gain, input resistance and output resistance of the common-source amplifier circuit in figure 1.

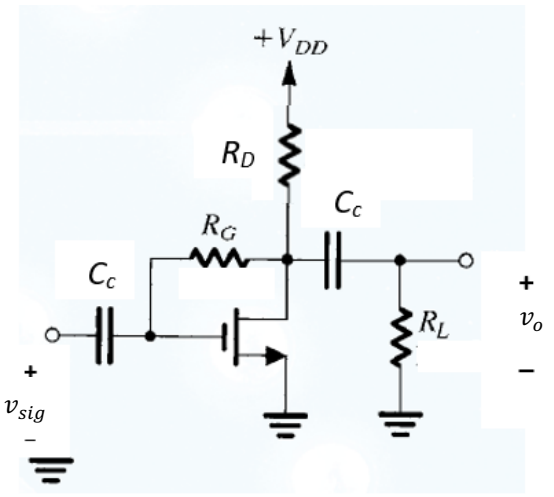


Figure 1: A Common-source Amplifier Circuit.

Solution

Equipment

Table 1. Equipment for experiment 7.

	Equipment	Model	Serial No.	S No.
1	BreadBoard Station			
2	Resistors 10 kΩ (2), 100 kΩ (1), 1 MΩ (1)			
3	NMOS transistor (1)	2N7000	https://www.onsemi.com/pub/Collateral/2N7000-D.PDF	
4	Capacitors 22 μF (2)			
5	Multi-meter (1)			
6	Probes and cables			

Experimental Procedure

Part (1): Amplifier Voltage Gain.

1. Connect the circuit in figure 1 without the ac source, capacitors and the load resistor. Take V_{DD}=10 V, R_D=10 kΩ and R_G=1 MΩ.

2. Measure the DC voltages at the drain and at the gate. Measure the voltage across R_D to find the DC drain current (Use $I_D = \frac{V_{R_D}}{R_D}$). Make sure the transistor is operating in saturation mode.
3. Now add the capacitors and the load resistor (R_L=10 kΩ) and connect a sinusoidal ac source v_{sig}. Start with a small amplitude (in mV) at 10 kHz for v_{sig}.
4. Connect both the input signal and the output terminals to the oscilloscope using two channels.
5. Observe both waveforms simultaneously and produce a sketch using graph 1.
6. Increase the amplitude of v_{sig} in steps of 10 mV. In each case record the amplitudes of v_{sig} and v_o in table 2.
7. Continue until you notice that the output becomes distorted. Mark the maximum input and output before distortion.

Part (2): Input and Output Resistance of Amplifier.

1. Reconsider the circuit in figure 1. Set the input signal at a comfortable value within the linear range of the amplifier.
2. To find the output resistance of the amplifier, measure the voltage gain with and without the load resistor. Record results in table 3.
3. To find the input resistance, connect a resistor (R_x=100 kΩ) between the input voltage source and the coupling capacitor (in series). Measure the voltage gain with and without R_x. Record results in table 3.

Results and Analysis

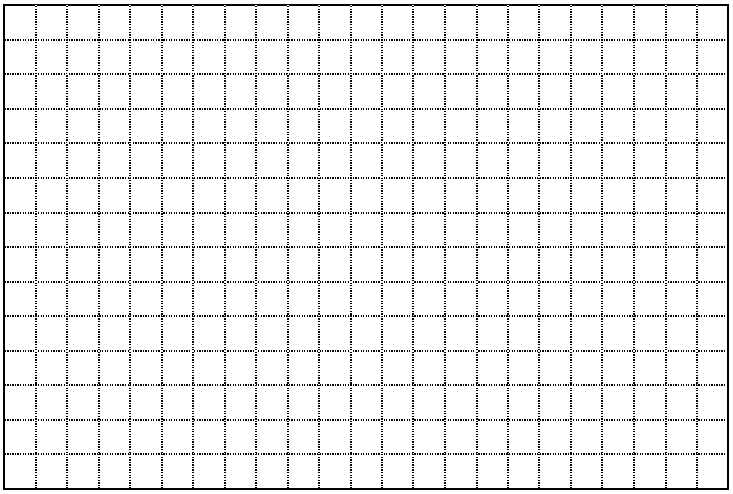
Part (1): Amplifier Voltage Gain.

1. Calculate the voltage gain of this amplifier (v_o/v_{sig}) and record your results in table 2.
2. Comment on the linearity of the amplifier in this circuit.

Answer:

3. What is the maximum output swing for this set-up of the amplifier? Explain.

Answer:



Graph 1: Amplifier input/output waveforms.

Table 2. Voltage Gain Results

Amplitudes		
v_{sig} (mV)	v_o (mV)	Gain

Part (2): Input and Output Resistance of Amplifier.

1. Calculate the input and output resistances of this amplifier circuit and record them in table 3.

Table 3. Input/output Resistances.

Output Resistance (kΩ)			Input Resistance (kΩ)		
Gain with R_L	Gain without R_L	R_o	Gain with R_x	Gain without R_x	R_i

2. Comment on your results based on your knowledge of the properties of the common-source amplifier.

Answer:

LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.

Name		Name		Name		Name	
ID		ID		ID		ID	
Section							